



Influence of Data Augmentation Strategies on the Segmentation of Oral Histological Images Using Fully Convolutional Neural Networks

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Abstract

Segmentation of tumor regions in H & E-stained slides is an important task for a pathologist while diagnosing different types of cancer, including oral squamous cell carcinoma (OSCC). Histological image segmentation is often constrained by the availability of labeled training data since labeling histological images is a highly skilled, complex, and time-consuming task. Thus, data augmentation strategies become essential to train convolutional neural networks models to overcome the overfitting problem when only a few training samples are available. This paper proposes a new data augmentation strategy, named Random Composition Augmentation (RCAug), to train fully convolutional networks (FCN) to segment OSCC tumor regions in H & E-stained histological images. Given the input image and their corresponding label, a pipeline with a random composition of geometric, distortion, color transfer, and generative image transformations is executed on the fly. Experimental evaluations were performed using an FCN-based method to segment OSCC regions through a set of different data augmentation transformations. By using RCAug, we improved the FCN-based segmentation method from 0.51 to 0.81 of intersection-over-union (IOU) in a whole slide image dataset and from 0.65 to 0.69 of IOU in a tissue microarray images dataset.

Keywords Data augmentation · Oral tumor segmentation · Fully convolutional neural networks · H& E-histological image · Whole slide images · Tissue microarray images

Introduction

According to the World Health Organization (WHO), cancer is the second leading cause of death in the world, accounting for 18.1 million cases and 9.6 million deaths in 2018 [1]. Cancer cases continue to grow globally and by 2030 the WHO estimates 24.1 million new cancer cases worldwide [2]. In Brazil, projections from the National Cancer Institute (INCA) show that 626,000 new cases of cancer will be registered in 2021, including 15,000 new cases of oral cancer [3]. The majority of oral cavity-derived cancers are oral squamous cell carcinoma (OSCC) [4]. The main risk factors for OSCC development are tobacco use, including smokeless tobacco, human papillomavirus infection, and alcohol drinking [5–7].

OSCC is among the 10 most frequent cancers affecting men and women, and among the 20 leading causes of death in the population [8]. It has a poor prognosis as most of them are diagnosed in advanced stages (III and IV); hence, it has a low survival rate and high-cost treatment [9]. Compared with other tumor locations, the treatment of OSCC patients is highly associated with comorbidities including speech problems, oral pain, malnutrition, sexual problems, dysphagia, and loss of appetite, thus leading to poor health-related quality of life [10]. It is in this context that OSCC can be considered one of the main public health problems in the world, and efforts must be made to facilitate the diagnosis, as well as to discover the mechanisms that drive its development and progression.

Cancer diagnoses are made in several steps including clinical examination, incisional biopsy, histopathological analysis through the microscope, and tumor grading [11, 12]. In addition, cellular characteristics and tissue components are also important information with diagnostic and prognostic implications [13]. Morphological changes in the size and shape of tumor cell nuclei are some of the basic features used to diagnose any kind of cancer [14]. The presence

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of keratinization and keratin pearls is also investigated by pathologists to determine OSCC grading, while tumor-infiltration lymphocytes are commonly analyzed to establish prognoses [12, 15]. This histological aspect in malignant cells is recognizably different from those observed in non-malignant cells. This microscopic investigation method, despite being referred to as the gold standard in cancer diagnosis, is influenced by inter- and intra-observer variability and is a time-consuming task that requires well-trained and experienced pathologists [4].

The increase of computational power in recent years, the reduction in the cost of image acquisition devices, and the pattern recognition techniques evolution have allowed computer-aided diagnosis (CAD) to become important nowadays. Fast scanners can be used to capture tissue samples into digital images to obtain the so-called digital histological images, allowing to view tissue samples on a computer rather than through a microscope. From the histological images, the use of CAD systems can be applied as a key tool to aid and enhance the pathologists' decisions, minimizing human interventions, discovering measurable and traceable cell and tissue information of significance prognosis, and reducing inter- and intra-observer variability [8].

Segmentation of tumor regions in hematoxylin and eosin (H &E)-stained histological images is an important task for a pathologist while diagnosing different types of cancer, including OSCC. The initial computational proposals for histological image segmentation were based on clustering [16], morphological operations [17], and thresholding techniques [18]. These handcrafted methods have some limitations when applied in histological images, as many parameters have to be set empirically, which will not produce consistent results if there is considerable variability within the image sets [19]. The results achieved by a deep learning model known as AlexNet [20] provided a major breakthrough in the computer vision community. These techniques based on convolutional neural networks (CNN) and fully convolutional neural networks (FCN) also have been successfully employed in medical and biological research [21].

Deep learning techniques traditionally need large amounts of labeled data to be trained. Unfortunately, histological images segmentation is often constrained by the availability of labeled training data since labeling histological images is a highly skilled, complex, and time-consuming task. One way to surpass this drawback is data augmentation. Data augmentation encompasses a set of data-space transformation techniques that enhance the size and quality of training datasets, hence deep learning models can be built using them [22]. In this paper, we address the challenging problem of automatically identifying and segmenting tumor regions of interest (ROI) on H &E-stained OSCC histological images. The proposed methodology uses an FCN to segment image patch-based oral tumor regions. Experiments

were also conducted to evaluate different data augmentation transformations, including a proposed augmentation using a generative adversarial network (GAN) [23] image inpainting transformation, to train fully convolutional networks to segment oral tumor regions in H &E-stained slides. Given the input image and their corresponding label, we propose Random Composition Augmentation (RCAug), a data augmentation pipeline composed of a random composition of geometric, distortion, color transfer, and generative image transformations are executed on the fly. The segmentation method (including the different augmentation strategies) was evaluated using two different datasets of H &E-stained OSCC histological images and the results were compared with state-of-the-art methods [24].

Related Work

In recent years, deep learning techniques have been proposed to address a wide variety of applications, such as the segmentation of tumor regions and cellular structures of interest. Most studies are conducted mainly on the most common tumors (i.e., prostate or breast cancer) while OSCC using whole slide images (WSI) or tissue microarray (TMA) images has been scarcely studied [25]. There are a few previous papers addressing automated methods to segment OSCC tumor regions using H &E-stained histological images. However, the vast majority of the existing approaches use CNN-based methods to classify small image-patches between tumor and normal patches to afterward rebuild a segmented version of the original images [15, 26, 27].

Santos et al. [28] propose to use an FCN-based model to localize and perform refined segmentation of OSCC regions in H &E-stained histological WSIs. The authors proposed a color-based tissue detector in a preprocessing step to remove background and scanning artifacts and an FCN model to localize and segment 640×640 image-patch-based tumor regions. To train and validate the method, a proprietary WSI dataset of H &E-stained tissues from OSCC regions was employed. As none similar OSCC labeled dataset was publicly available at the time, for comparison, the authors tested the proposed segmentation method on a publicly available WSI dataset from breast cancer-derived lymph node metastases—CAMELYON16 [29]. The influence of different color spaces (CIE L*a*b*, HSV and RGB) and different image-patch sizes (160×160, 320×320 and 640×640) were also evaluated. The CIE L*a*b* color space and the larger image-patch size (640×640 pixels) achieved the best results with 0.85 of mean intersection-over-union (IOU) in the proprietary dataset and 0.84 of IOU in the CAMELYON16 dataset.

Martino et al. [24] recently published the ORal Cancer Annotated (ORCA) dataset. ORCA is a labeled dataset

containing 200 H &E-stained histological OSCC images from The Cancer Genome Atlas (TCGA) dataset [30]. To the best of our knowledge, this is the first publicly available segmentation-proper labeled dataset containing OSCC tumor sample images. The authors used the ORCA dataset to compare the segmentation results achieved by four different FCN-based architectures inspired by the U-Net model. The segmentation performance was measured using a down-sampled version of the original image. All the original input images were resized to 512×512 pixels before being segmented by the FCN models into three different class labels: carcinoma pixels; tissue pixels not belonging to a carcinoma; non-tissue pixels. The authors also evaluated the use of the red channel of the RGB color space in combination with the hue channel of the HSV color space. The FCN segmentation model with the best configuration achieved 0.56 of IOU for the carcinoma class.

Although segmentation is the most common subject of papers applying deep learning to medical imaging [31], to the best of our knowledge, there are only a few papers to investigate OSCC detection on WSI or TMA images applying FCN-based methods [8, 21, 25]. Moreover, even though data augmentation is commonly used, it is not easy to find studies evaluating the influence of data augmentation strategies based on the random composition of geometric, distortion, color transfer, and generative image transformations to segment WSI or TMA images using FCN.

Materials and Methods

Image Datasets

Two different public datasets of H &E-stained histological images were considered: the ORCA [24, 30] and OCDC [32] datasets. Both datasets contain digitized images from OSCC tumor samples. The ORCA dataset was used to evaluate the proposed segmentation method and the proposed data augmentation strategy on a publicly available dataset of TMA images with different resolutions. TMA is a tool to extract small tissue cores from regions of interest from paraffin-embedded tissues. Unlike a WSI, where the entire histological slide is digitized into a large single image, TMA images represent small pieces of tumor regions of interest organized into a microarray. The OCDC dataset was used to evaluate the proposed RCAug on WSIs obtained from OSCC samples.

The original images of both datasets—ORCA (TMA) and OCDC (WSI)—were employed to extract image patches of 640×640 pixels without any resize or resolution modifications from the original images. In addition, ORCA dataset was also evaluated using resized images of 512×512 pixels. In both datasets we can find patches with tumor regions

and patches of normal tissues like the serous salivary gland, connective tissue, mucous salivary gland, striated muscle, keratinized epithelial tissue, and oral mucosa. A detailed description of each dataset is presented in the following. Table 1 summarizes information from the image datasets.

Oral Cancer Annotated (ORCA) Dataset

ORCA is a public labeled dataset containing 200 TMA images from representative OSCC tumor areas extracted from WSIs of the TCGA dataset [24, 30]. The 200 labeled images were split into 100 images to train and 100 to test and contain images with two different dimensions: 4500×4500 or 2250×2250 pixels. The ground-truth data for training and testing are provided as mask images containing three different class labels: carcinoma pixels (white), tissue pixels not belonging to carcinoma (gray), and non-tissue pixels (black). In our experiments, a training dataset was built using 4181 image-patches of size 640×640 pixels extracted from the original 100 images of the training dataset. The testing dataset was built using 4481 image-patches extracted from the 100 original images of the testing dataset. Figure 1 illustrates an original image from the ORCA dataset, some image-patches extracted, and their corresponding ground-truth.

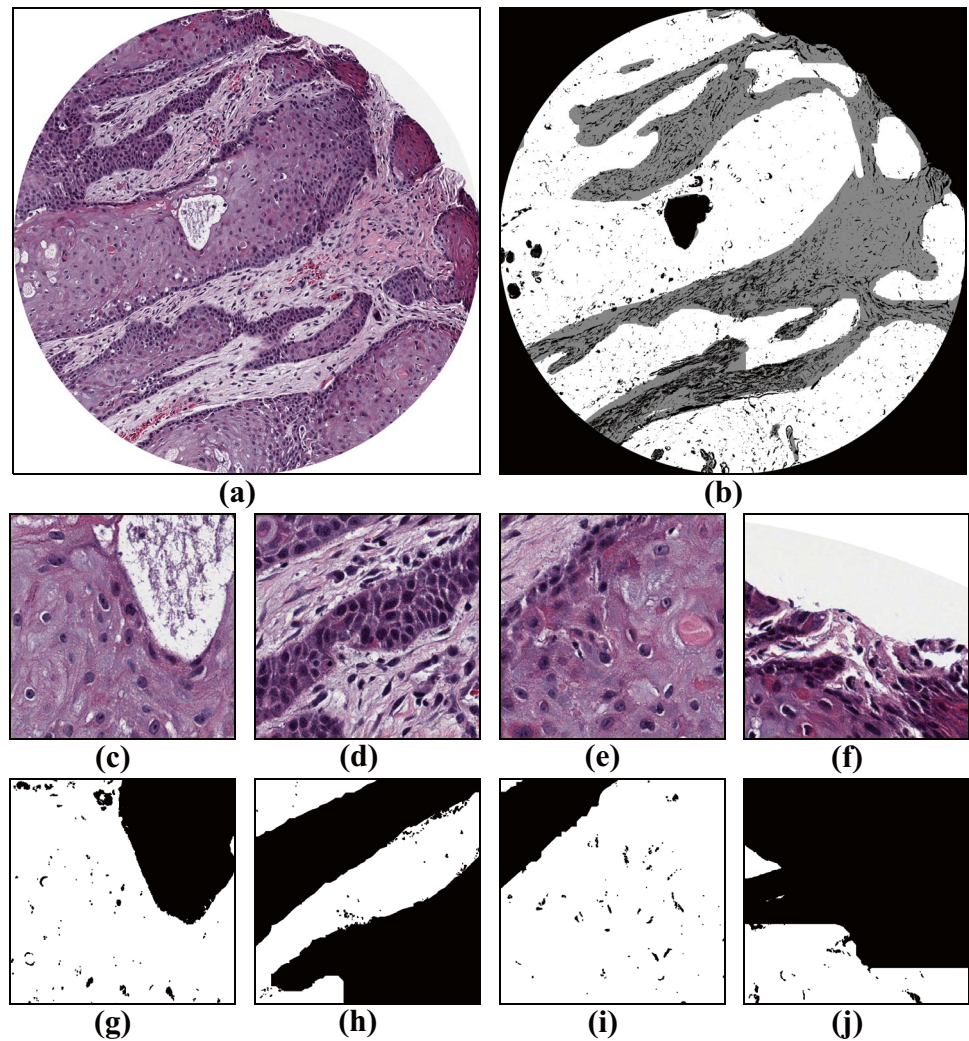
The 200 original ORCA TMA images were also used to build a derived dataset in the same way as published by ORCA dataset authors. The 100 images from the training dataset and the 100 images from the testing dataset were resized to 512×512 pixels to compose the dataset identified as ORCA 512×512. This dataset was chosen for external validation and used in the same way as published by Martino et al. [24].

Oral Cavity-Derived Cancer (OCDC) Dataset

The H &E-stained OCDC dataset is a public dataset that consists of tissue images and their corresponding tumor region annotation masks [28, 32]. The OCDC dataset was built using tissue specimens retrieved from the Department of Oral and Medical Pathology Services archives in the Federal University of Uberlândia between 2006 and 2013. All the tissue specimens were collected from human patients diagnosed with OSCC under approval by the Committee on Research and Ethics of the institution.

To create the OCDC dataset, tissue samples obtained after surgical procedure in 15 OSCC-affected patients were digitized into 15 WSIs using the Slide Scanner Aperio AT2 (Leica Biosystems Imaging, Inc., Nussloch, Germany) coupled to a computer (Dell Precision T3600) at ×20 magnification and pixel-level resolution of 0.5025 μm × 0.5025 μm . The digitized images have different sizes—the larger one has almost three billion pixels (63,743×45,472 pixels)—and were stored in SVS format using the RGB color space. A total of 1020 image-patches of

Fig. 1 Images from the ORCA dataset: 1st row (a–b) shows an original image and its corresponding ground-truth; 2nd row (c–f) shows image-patches (640×640 pixels) extracted from the original images; 3rd row (g–j) shows the annotated tumor regions for each corresponding image-patch from the 2nd row



size 640×640 pixels were randomly extracted from these 15 WSIs and the content of each image-patch was hand-annotated by a pathologist. This kind of task and a large amount of information present in the images require very skilled professionals and a great effort to label the relevant structures to create this kind of image datasets.

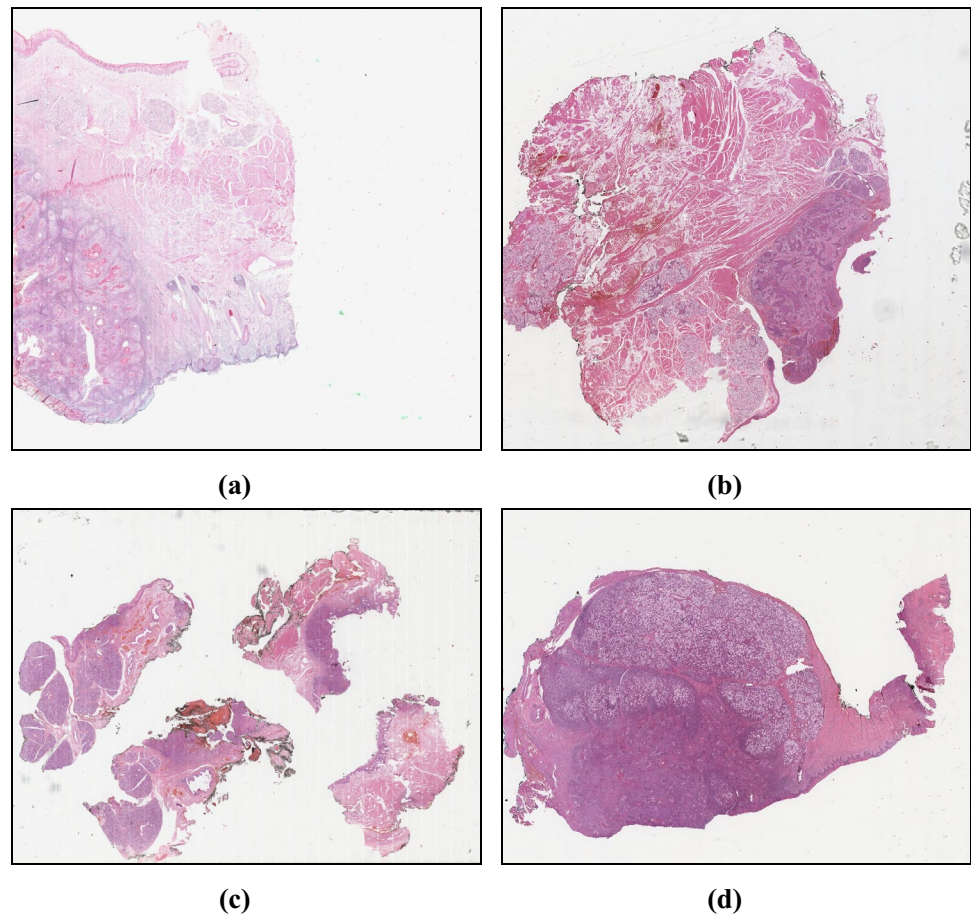
The image-patch dataset was split into two subsets: the training set with 840 image-patches and the test set with 180 image-patches. WSI samples from the OCDC dataset are illustrated in Fig. 2. Representative examples from the 1020 produced image-patches are shown in Fig. 3. Figure 3(m–r) also shows the corresponding hand-annotated regions produced from the image-patches presented in Fig. 3(g–l).

Segmentation Method

The method used for segmenting tumor regions in oral histological tissue images is the one originally proposed by Santos et al. [28]. This method is established on the

following major steps: (1) the original histological image is scaled down at a factor of 32× to perform background removal and tissue detection using a color-based tissue region detector; (2) the tissue regions identified on the scaled-down image are mapped back into the original histological image; (3) image-patches of size 640×640 pixels are extracted from the original histological images on the original magnification considering only the identified tissue regions; (4) a pixel-level cancer probability (a value between 0.0 and 1.0) is computed using an FCN model; (5) the final tumor region segmentation is achieved by thresholding the final probability map at 0.5. The FCN architecture, which consists of a couple of encoding (down) and decoding (up) layers, is based on U-Net model [33, 34] and outputs a gray-level image with the cancer probability for the input image—a heatmap using a blue-to-red color map representation is used for better visualization of the probabilities. Figure 4 illustrates an overview of the segmentation method. All the model hyperparameters

Fig. 2 Four WSIs from OCDC dataset



were used as specified by Santos et al. [28] without any adjustment modifications.

The segmentation method was previously evaluated using WSI datasets with images of different diseases: OSCC and breast cancer-derived lymph node metastases [28]. The WSIs digitize the whole histological slide in large images created from tissue sections that can contain many different types of cells and regions—such as keratin, lymphocytes, blood vessels, glands, muscle, and tumor cells. In this paper, TMA images representing small pieces of previously selected tumor regions and the impacts of using data augmentation to train fully convolutional segmentation models are also evaluated. For datasets in which the full images are smaller than 640×640 pixels, as the case of the ORCA 512×512 dataset, the preprocessing stage is not necessary (see Fig. 4).

Random Composition Augmentation (RCAug)

There are several data augmentation techniques and different approaches can be more or less suitable when applied to specific problems. For example, geometric transformations such as rotation and translation of images could be very useful to

obtain an augmented dataset version that avoids positional biases present in the training data. On the other hand, rotation and flip operations could be not safe or suitable, as, for example, in digit recognition tasks such as number 6 versus number 9. Most of the existing data augmentation operations focus on improving image classification models. For segmentation task, there is an extra effort to address, once data augmentation operations typically need to be done with transformations that are applied to both the images and their corresponding labels equally [22].

Standard data augmentation approaches consist of combining transformations to generate augmented data before training the model. For instance, geometric transformations are applied to create a larger dataset; thus, for each original image present in the dataset, a new image is generated after applying flip, rotation, distortion, and other operations. For a dataset of size N and T different transformations, an extended dataset of size $N \times T$ is generated. The extended image dataset containing the augmented and original images is then fed into the model at the training stage.

In this paper, we propose a data augmentation strategy that is based on a random composition of different transformations applied to the input images on the fly. Given the input

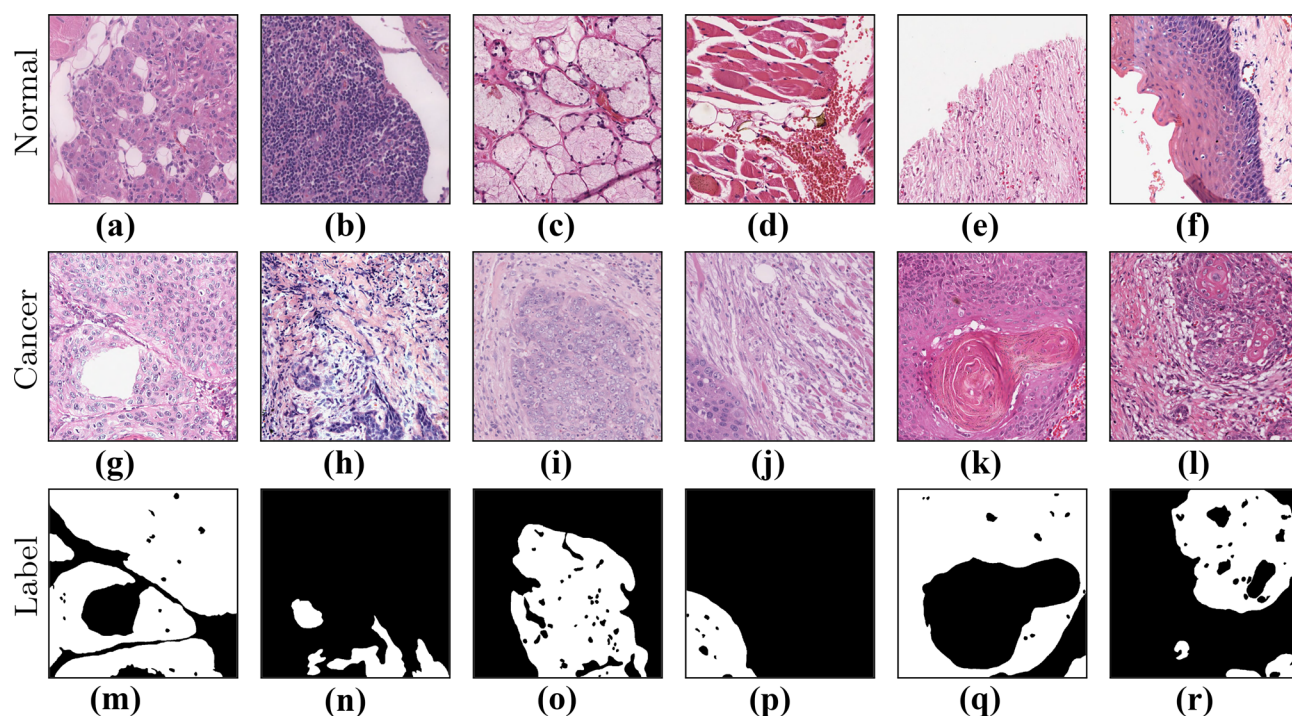


Fig. 3 Some image-patches (640×640 pixels) from the OCDC dataset that shows the anatomic diversity present in H &E-stained histological images: 1st row (a–f) shows normal regions (serous and mucous salivary gland lobules, tonsil, striated muscle, and connective and epi-

thelial tissues); 2nd row (g–l) shows OSCC regions with and without keratin pearl-containing areas; 3rd row (m–r) shows the produced hand-annotated tumor regions (in white) by a pathologist

image and their corresponding label, a pipeline is executed with different geometric, distortion, color transfer, and generative image transformations. At each pipeline step, a roll dice with a probability of applying the transform equal to 0.5 is performed to choose if the corresponding transformation will or will not be executed. This parameter was chosen empirically on the investigated histological image datasets. After each step, the resulting image and their corresponding label—that may or may not have been augmented—are then used as the input to the next pipeline step.

The proposed strategy executes image augmentations on-the-fly during the model training and does not generate the augmented training image dataset in advance. This provides a much more flexible manner to increase the training dataset size. Since image datasets to train deep learning models are typically large in size, this approach allows overcoming the limitations of the storage space without penalizing the diversity and amount of image transformations. This strategy also generates an undefined amount of new training images using unique arrangements from different augmentation transformations on the fly. A total of eight transformations were used: horizontal flip, vertical flip, rotation, transpose, grid distortion, optical distortion, color transfer, and inpainting. Each used image transformation is detailed in Sect. 3.3.1 and Fig. 5 illustrates an overview of the proposed data augmentation strategy.

Data Augmentation Transformations

Digitized images from histological slides must be invariant to positioning and orientation. In this way, for histological images, geometric transformations are crucial to train deep learning models that avoid positional biases present in the training data. Furthermore, geometric image transformations are some of the most common and easiest transformations to implement. Four different augmentations based on geometric transformations were used: rotation, transpose, and horizontal and vertical axis flipping [35]. These image transformations are applied to both images and their corresponding labels. Figure 6(c/i), (d/j), (e/k), and (f/l) exemplifies the geometric transformations used as data augmentation.

Tissue folding, image distortions, blurring, and even color differences caused by the staining process and lightning conditions are common undesired artifacts that reduce the quality of the histological images. To minimize problems related to distortions and color biases, augmentation transformations

Table 1 Summary of the image datasets used for the training and test of the segmentation method

ORCA	200	4481	4181	100
ORCA 512×512	200	100	100	400
OCDC	15	210	840	400

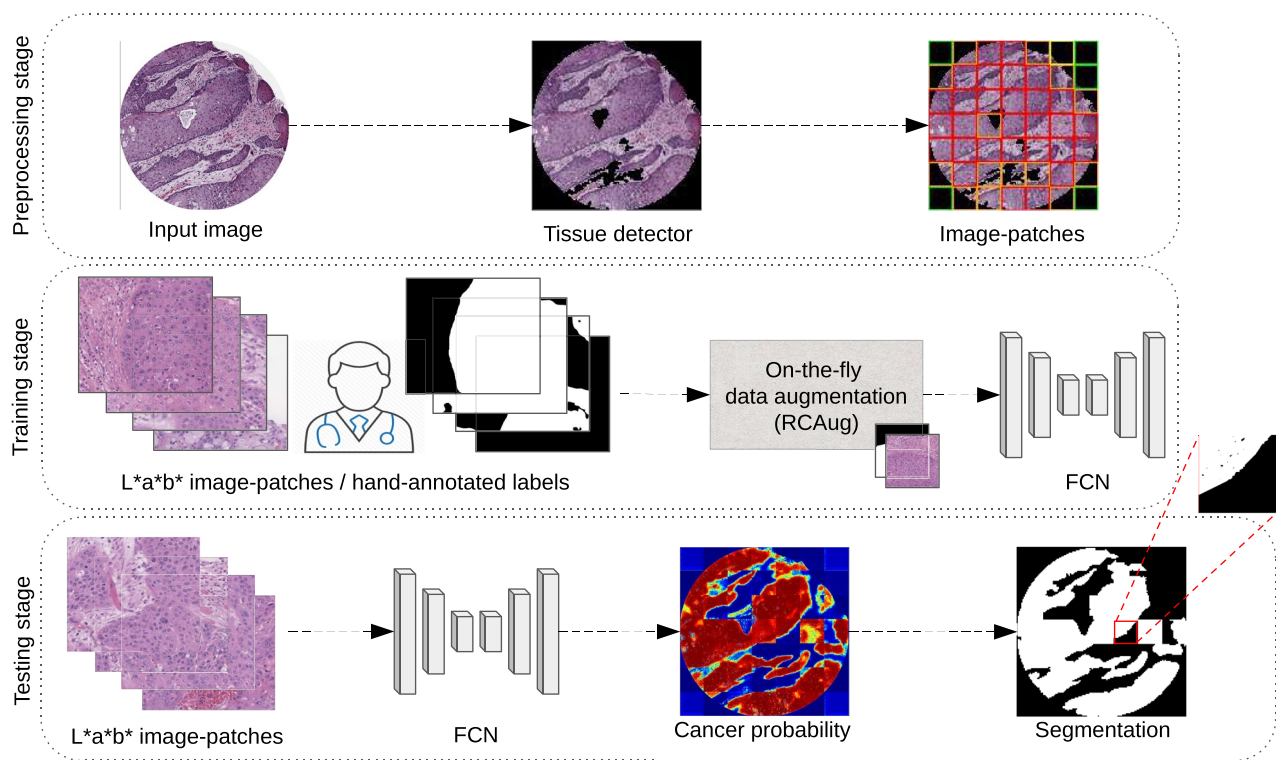


Fig. 4 Overview of the segmentation method. Preprocessing stage: the identified tissue regions of the histological images are split into image-patches of size 640×640 pixels. Training stage: a patch-based FCN is trained using data augmentation operations. Testing stage: the cancer probability heatmap is computed using the trained FCN;

the tumor regions are localized and segmented by thresholding the heatmap image. An image-patch sample is highlighted to show that the segmentation method is able to obtain a refined segmentation of tumor regions [28]

considering grid distortions, optical distortions, and color appearances were used. The distortion transformations used as data augmentation were implemented using the Albumentation [35] open-source Python library and are illustrated in Fig. 6(g/m) and (h/n). The used color augmentation transformation was proposed by Xiao et al. [36]. This technique combines the structure of one image with the color appearance of another image to construct augmented images on-the-fly for each training iteration. More specifically, for each training image, a target image is randomly selected from the

training set and a new augmented image is generated through a color transfer procedure. Using the color transfer augmentation, the augmented image takes the color appearance from the target image while its structure remains the same as in the original image [36]. Figure 7a illustrates the used color transfer transformation.

To make the training process more robust to noise and occlusion problems, a combination of random erasing [37] with image inpainting [38] transformations is also proposed as data augmentation. Random erasing was implemented by

Fig. 5 Random Composition Augmentation (RCAug): before each image and label be used for training, a random composition of eight transformations is defined on the fly. In this example only the transformations highlighted in red were randomly chosen and applied to the input image and their corresponding label: Rotation, Grid distortion, and Color transfer

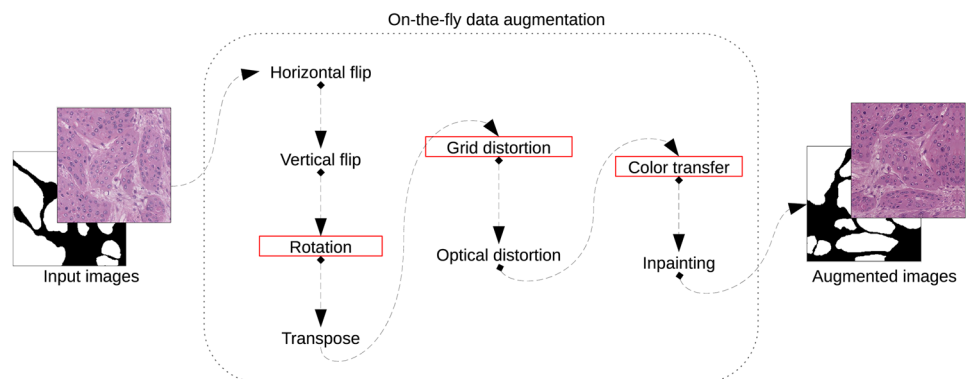
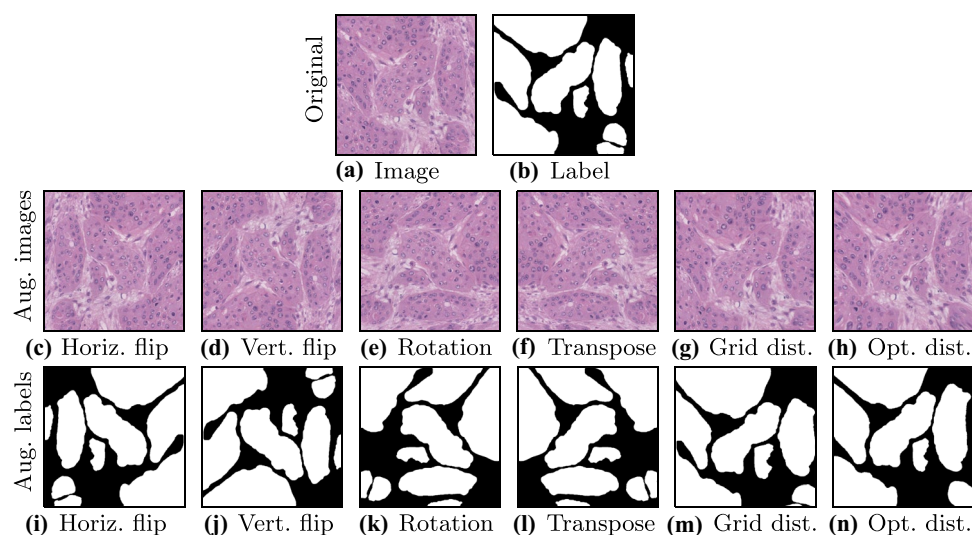


Fig. 6 Augmentation transformations applied to both the images and their corresponding labels equally: **a/b** original images; **c/i** horizontal flip; **d/j** vertical flip; **e/k** rotation; **f/l** transpose; **g/m** grid distortion; **h/n** optical distortion



randomly selecting a 128×128 pixels patch on the training image and using a GAN-based image inpainting transformation to restore the missing pixels of the selected patch. The size of 128×128 pixels was chosen because this is the largest size with good results that our available hardware supports. Figure 7b illustrates the proposed GAN-based image inpainting transformation. The used image inpainting technique was proposed by Yu et al. [38]. We use the Python implementation for the GAN-based image inpainting provided by the authors with the model pre-trained on ImageNet dataset [20] for 430k

iterations. All the augmentation transformations were used as specified by their respective authors with the provided default configurations. No adjustment modifications were applied.

Experimental Evaluation

To assess the quality of the method to segment OSCC tumor regions, experimental evaluations were performed using the images from the test subsets detailed in Sect. 3.1. The results

Fig. 7 The used augmentation transformations that do not need to be applied in the corresponding label images: **a** color transfer augmentation [36]; **b** GAN-based image inpainting augmentation [37, 38]. The original corresponding labels remain unmodified for these transformations

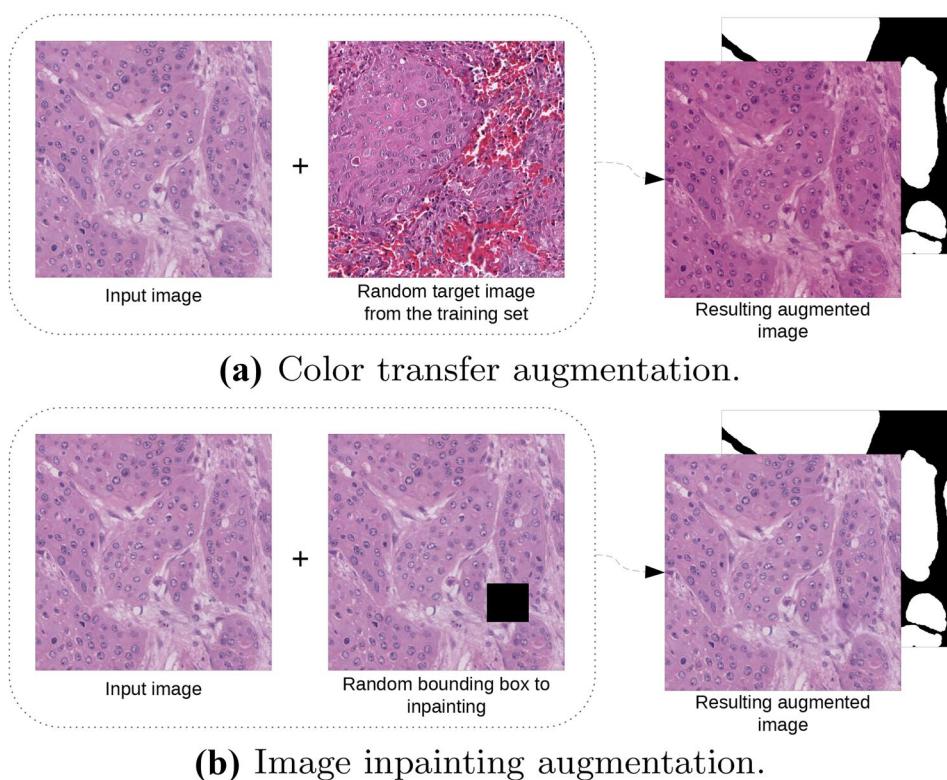
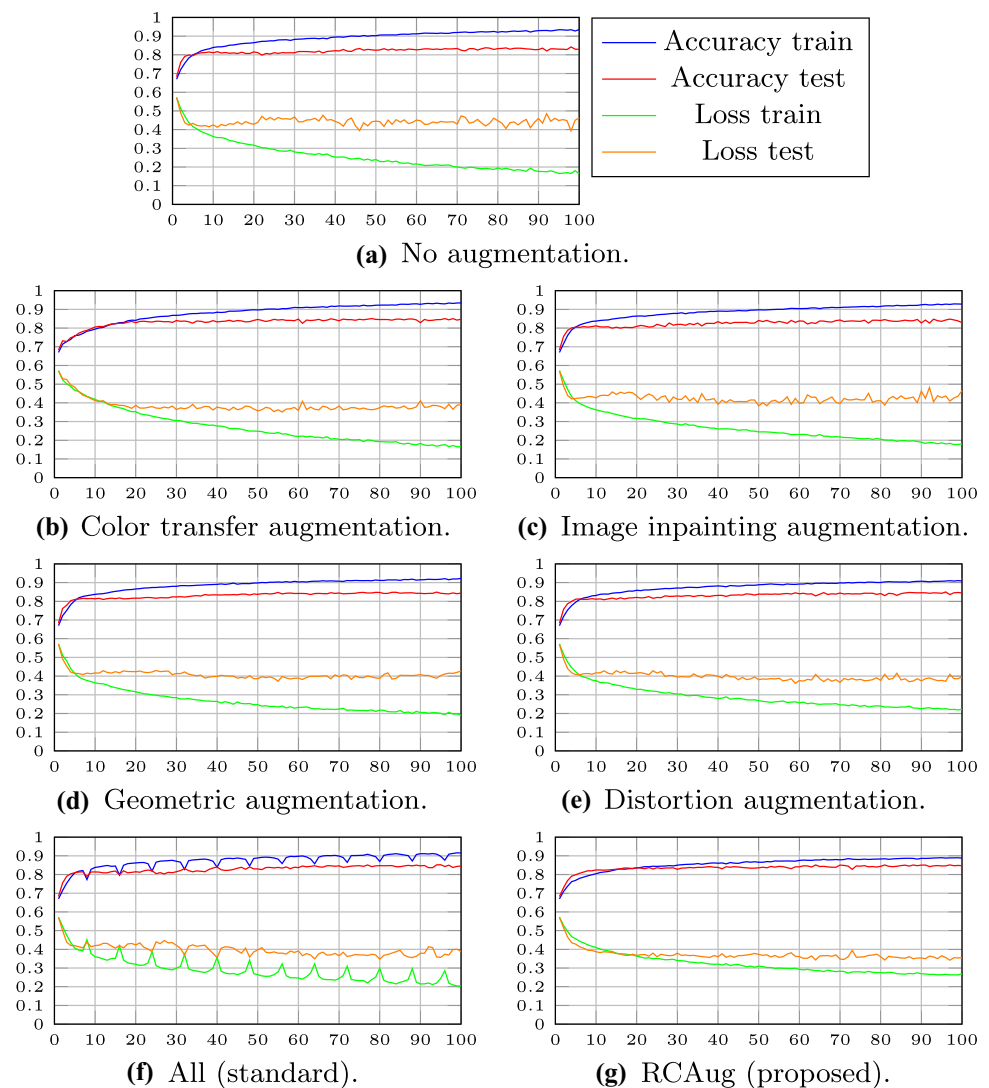


Fig. 8 Training/testing learning curves for accuracy and loss of the FCN-based segmentation model for the ORCA dataset. Smaller loss values over time indicate more learning and a value of 0.0 indicates that the dataset was learned perfectly and no mistakes were made



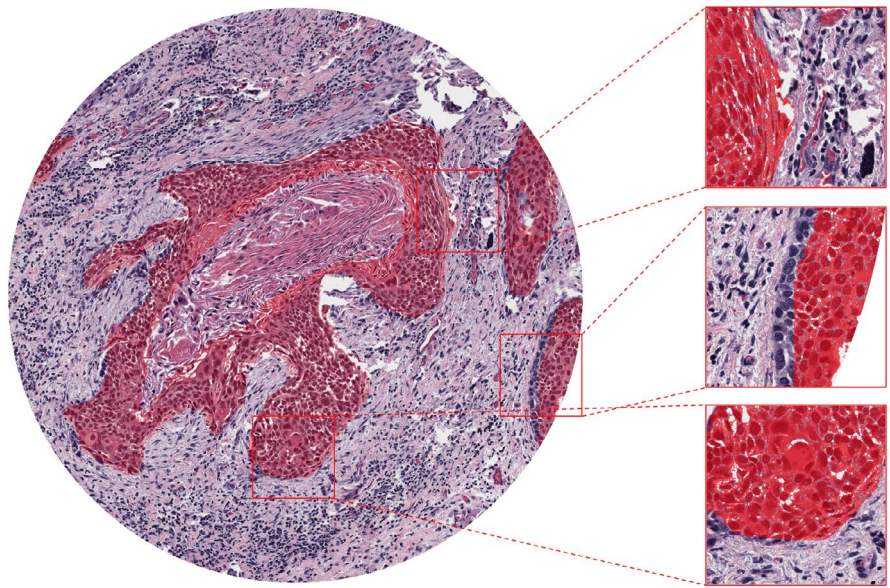
were compared quantitatively and qualitatively with the tumor regions manually segmented by the specialist. The localization and segmentation of tumor regions in the TMA or WSI images were visually analyzed by a specialist as a qualitative evaluation. The quantitative segmentation performance was measured by calculating the overlapping regions of the resulting automated segmented image and the regions of a reference image segmented by the specialist (gold-standard). The average performance was measured by some of the most commonly

used quantitative criteria concerning pixel-level classification: Accuracy, Precision, Sensitivity, F1-score (or Dice Coefficient), Intersection over union (IOU) (or Jaccard index), and Specificity. For the computation of these metrics, we considered as true positives (TP) the amount of correctly detected pixels, i.e., cancer detected as cancer; true negatives (TN) the amount of correctly undetected pixels, i.e. non-cancer detected as non-cancer; false positives (FP) the number of incorrectly detected pixels, i.e. non-cancer detected as cancer; and false

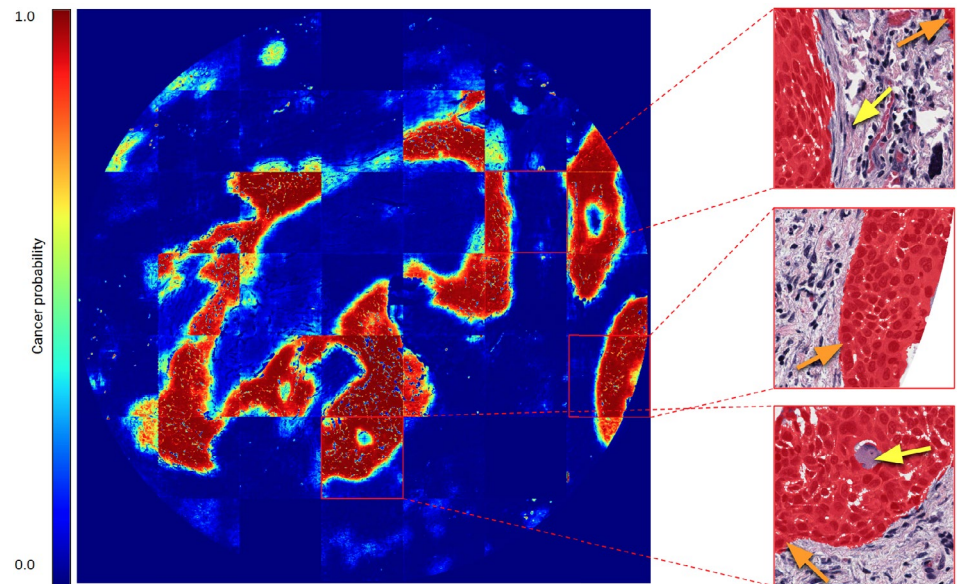
Table 2 Average quantitative results for test images from the ORCA dataset. The proposed RCAug data augmentation achieves the best results overall

Aug. transformation	Accuracy	Precision	F1	IOU	Sensitivity	Specificity
No augmentation	82.73%	78.75%	79.13%	65.46%	79.50%	84.99%
Color transfer	84.58%	82.27%	80.98%	68.04%	79.73%	87.98%
Image inpainting	82.87%	74.50%	81.01%	68.08%	88.77%	78.74%
Geometric	84.02%	81.10%	80.43%	67.26%	79.77%	86.99%
Distortion	84.44%	79.11%	81.73%	69.10%	84.53%	84.38%
All (standard)	84.42%	79.61%	81.53%	68.82%	83.54%	85.03%
RCAug (proposed)	84.60%	78.58%	82.14%	69.69%	86.04%	83.59%

Fig. 9 One of the TMA images from the ORCA dataset and three corresponding image patches extracted from the tumor areas. Identified tumor regions are highlighted in red; orange arrows indicate some false positive regions; yellow arrows indicate some false negative regions



(a) Original TMA image and three selected image-patches (the provided ground-truth regions are overlaid in red).



(b) Heatmap for tumor regions and three image-patch segmentation results for (a).

negatives (FN) the number of incorrectly undetected pixels, i.e. cancer detected as non-cancer [39–41].

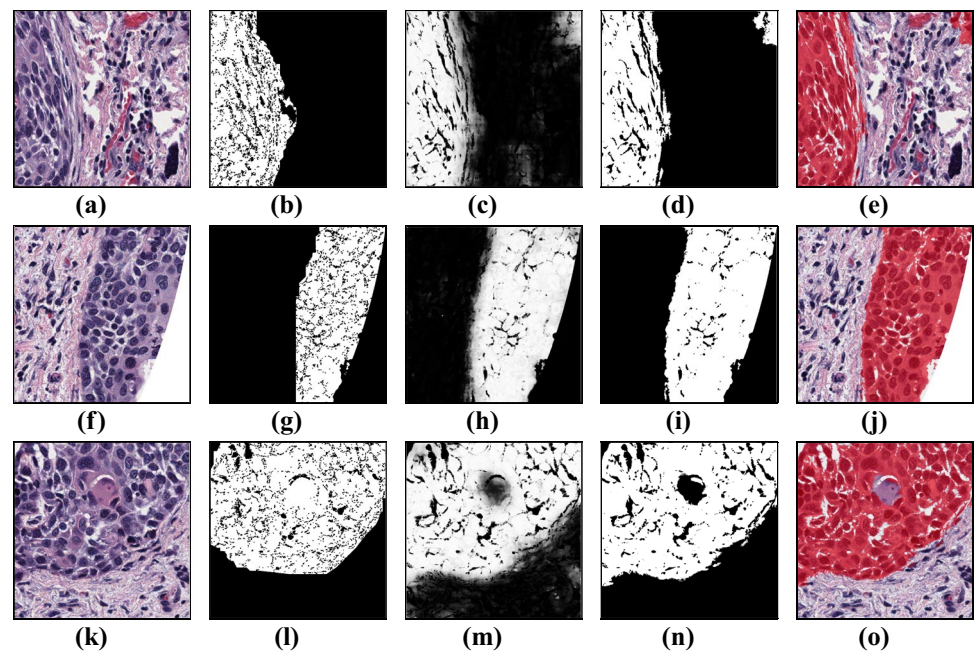
The FCN segmentation model was trained using the training subsets presented in Table 1. A separate training was performed for each dataset—ORCA, ORCA 512×512, and OCDC—using different settings of data augmentation transformations. The performance of the segmentation method was measured by evaluating the results using the images from the test sets.

The influence of different data augmentation transformations on the training of FCN models to segment OSCC tumor regions was also evaluated using all three datasets. The

following setup was evaluated: no augmentation; color transfer augmentation; image inpainting augmentation; geometric augmentation; distortion augmentation; standard augmentation using all transformations, and the proposed pipeline with random compositions of all transformations on the fly (RCAug, as described in Sect. 3.3). The FCN segmentation model was separately trained using each of these different configurations.

The segmentation method, the data augmentation transformations, and the experimental evaluations were implemented using the PyTorch framework [42] with the Albumentation [35] open-source Python library. The FCN

Fig. 10 Qualitative image-patch segmentation results for the ORCA dataset. First column **a/f/k** shows the input image-patches; second column **b/g/l** shows the gold-standard; third column **c/h/m** shows the output cancer probability for each input image; fourth column **d/i/n** shows the final tumor regions segmentation result; fifth column **e/j/o** shows the input image with the identified tumor regions overlaid in red



segmentation model was trained using a desktop computer (Intel Core i7 3.4GHz×8 processor, 32 GB memory, 1TB SSD) equipped with GeForce GTX 1050 Ti graphic card

and Ubuntu 20.04 operational system. The training time for the ORCA dataset during 100 epochs using 4181 images of size 640×640 pixels was about 7 days. For OCDC dataset,

Fig. 11 Training/testing learning curves for accuracy and loss of the FCN-based segmentation model for the ORCA 512×512 dataset

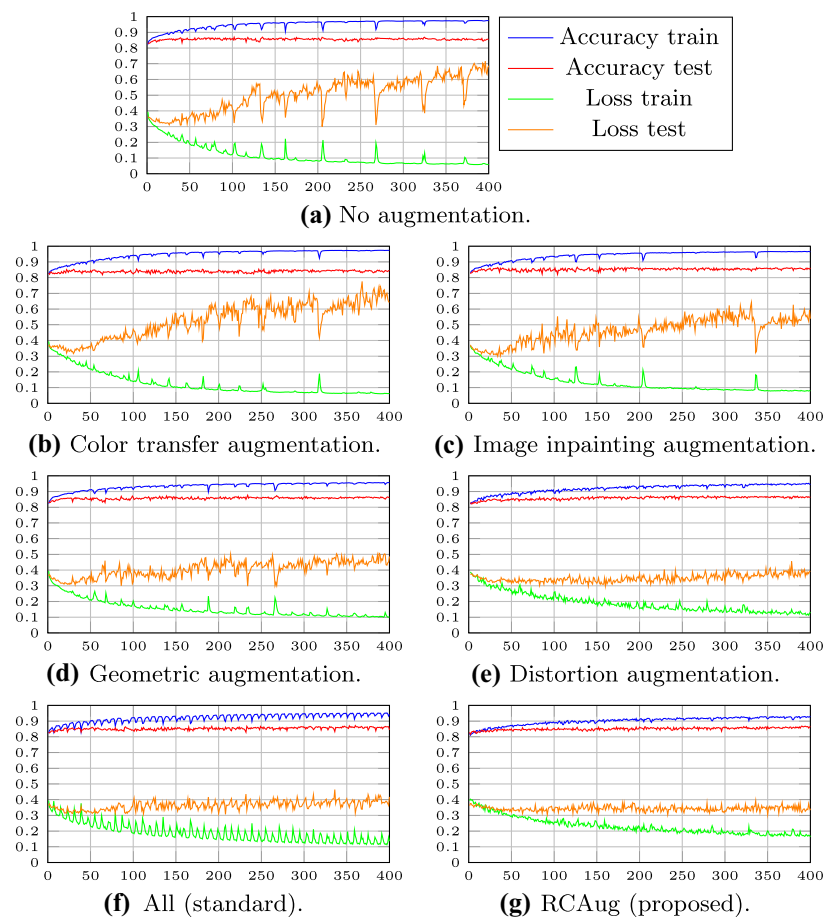


Table 3 Average quantitative results for test images from the ORCA 512×512 dataset

Aug. transformation	Accuracy	Precision	F1	IOU	Sensitivity	Specificity
No augmentation	85.45%	72.22%	75.45%	60.57%	78.97%	88.01%
Color transfer	84.62%	69.59%	74.88%	59.85%	81.04%	86.03%
Image inpainting	86.18%	74.41%	76.16%	61.50%	78.00%	89.42%
Geometric	85.71%	76.63%	73.83%	58.51%	71.22%	91.43%
Distortion	86.23%	73.93%	76.51%	61.96%	79.28%	88.97%
All (standard)	85.92%	76.45%	74.48%	59.34%	72.62%	91.17%
RCAug (proposed)	86.46%	73.81%	77.16%	62.81%	80.83%	88.68%

the elapsed time to train the model during 400 epochs was about 5 days using 840 images of size 640×640 pixels. For ORCA 512×512 we train the FCN model during 400 epochs in one day. After training, the elapsed time to process the input images was about 0.8 s for the images of 640×640 pixels size and 0.7 s for the 512×512 pixels size. To provide a better understanding and make this work as reproducible as possible, the source code is publicly available at https://github.com/dalifreire/tumor_regions_segmentation.

Results and Discussion

Figures 8, 11, and 13 show the model learning performance over time using different data augmentation transformations for each dataset: ORCA, ORCA 512×512, and OCDC. In these plots, we can depict the evolution training process with respect to the different strategies adopted. For each epoch, we can follow the evolution of the accuracy of the model considering the training and the test sets. Also, the

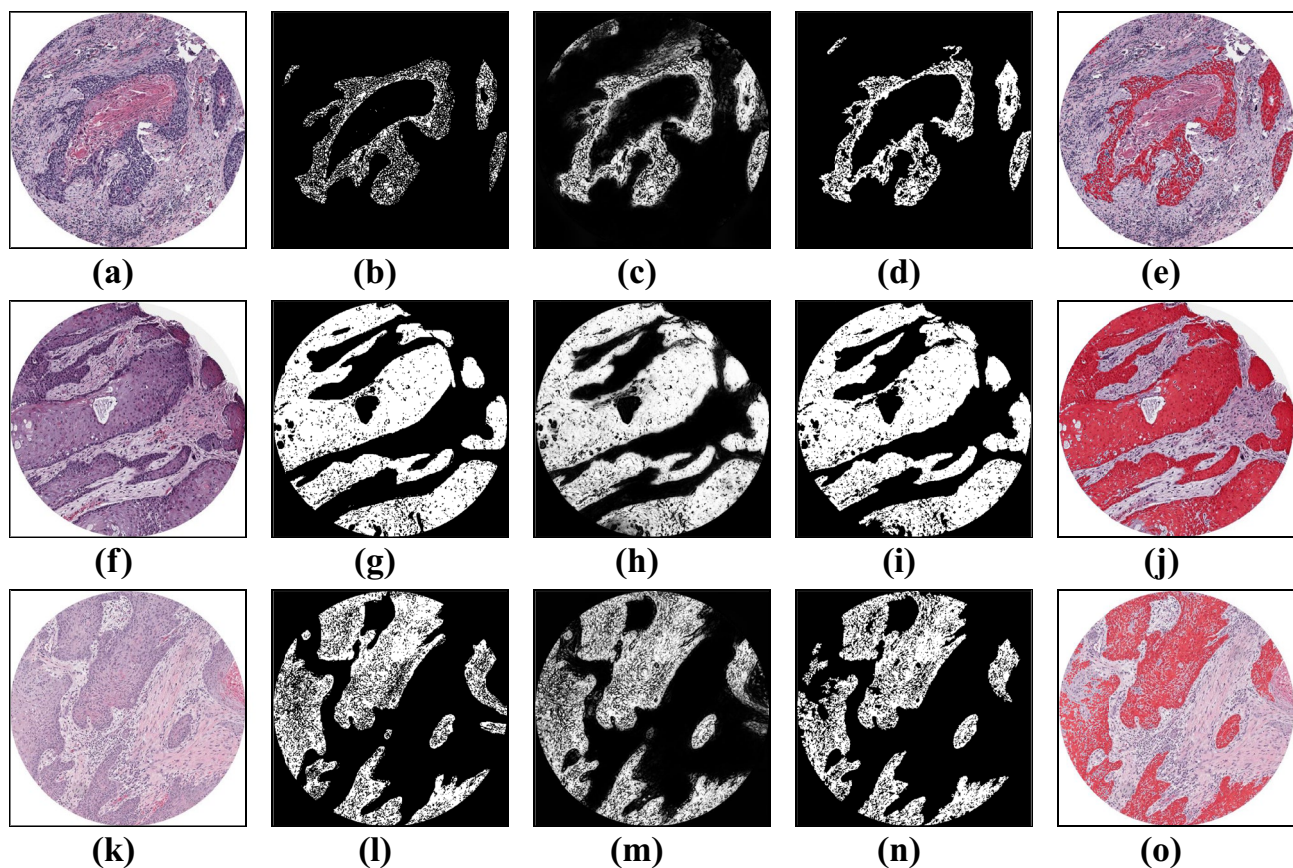
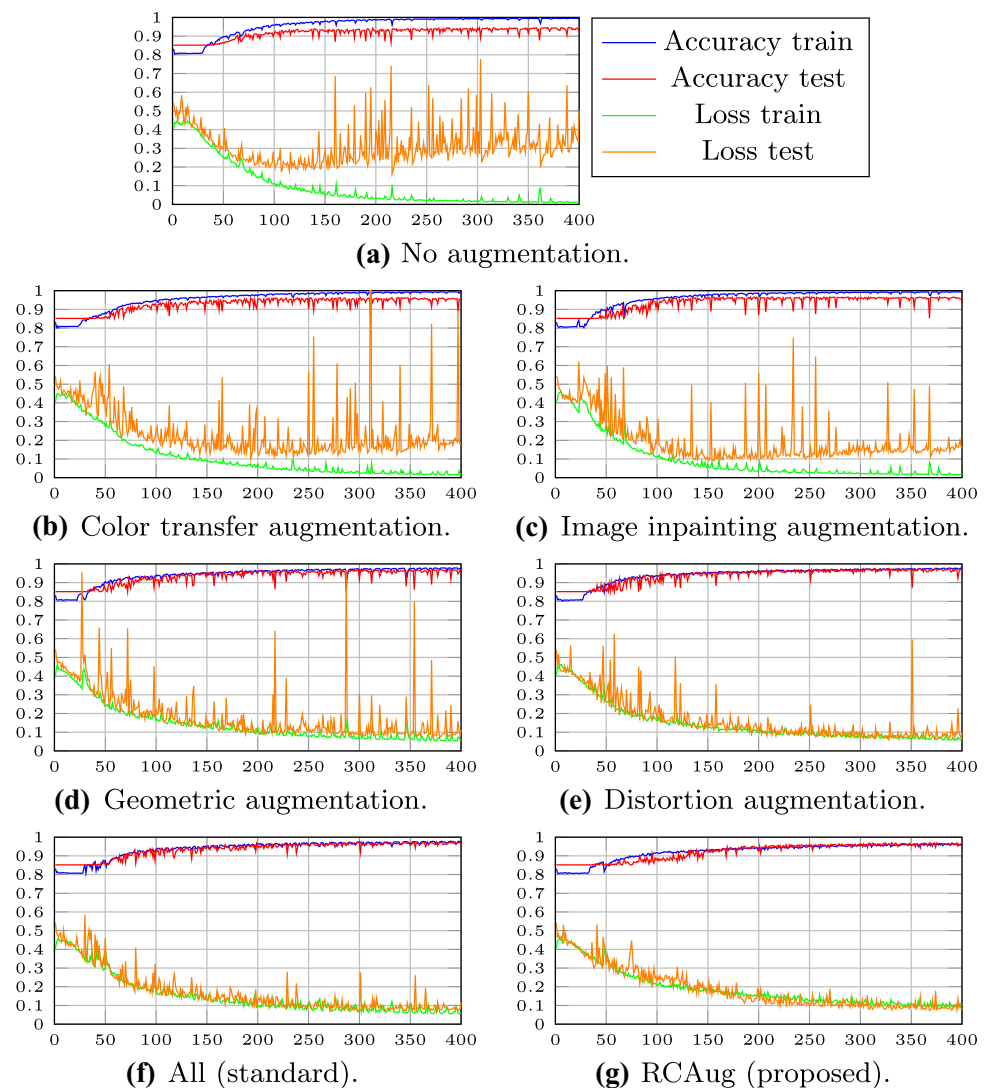


Fig. 12 Qualitative segmentation results for the ORCA 512×512 dataset. First column **a/f/k** shows the input image-patches; second column **b/g/l** shows the gold-standard; third column **c/h/m** shows the output

cancer probability for each input image; fourth column **d/i/n** shows the final tumor regions segmentation result; fifth column **e/j/o** shows the input image with the identified tumor regions overlaid in red

Fig. 13 Training/testing learning curves for accuracy and loss of the FCN-based segmentation model for the OCDC dataset



loss—i.e., how well the model is being trained—is also shown. For accuracy, it is expected that the model evolves to 1, while good values for loss are expected to be near 0. Tables 2, 3, and 4 summarize the average quantitative results achieved through the discussed data augmentation transformations in the ORCA, ORCA 512×512, and OCDC datasets. Friedman’s test was also applied to evaluate the statistical relevance between the methods considering the F-1 values. The p -value computed was 0.0348, which indicates statistically significant differences between the methods [43].

Figures 9 and 10 show comparative results between the gold standard and those achieved by the segmentation method, trained using our proposed data augmentation strategy in the ORCA dataset, for a qualitative analysis both in the whole TMA image and the image patches. Figure 12 shows some qualitative segmentation results for the ORCA 512×512 dataset.

From Tables 2, 3, and 4, we can see that the RCAug strategy is able to generally improve the segmentation results in all three evaluated datasets, as indicated in bold. The learning curves depicted in Figs. 8, 11, and 13 also reveal that our proposed data augmentation strategy, randomly composing different augmentation transformations, turns the training process more stable, reducing the gap between the training and test curves.

The experimentation results show also that each augmentation transformation—color transformation, GAN-based generative image inpainting transformation, geometric transformations, and distortion transformations—even when used separately improves the segmentation results (compared when no augmentation transformation is applied). However, as observed in the learning curves, the color transformation and the GAN-based image inpainting transformation had less influence on the segmentation

Table 4 Average quantitative results for test images from the OCDC dataset

Aug. transformation	Accuracy	Precision	F1	IOU	Sensitivity	Specificity
No augmentation	92.44%	89.83%	68.41%	51.98%	55.24%	98.91%
Color transfer	95.50%	90.70%	83.65%	71.90%	77.62%	98.61%
Image inpainting	96.36%	90.54%	87.29%	77.44%	84.27%	98.47%
Geometric	96.43%	86.38%	88.21%	78.91%	90.13%	97.53%
Distortion	96.70%	89.49%	88.78%	79.82%	88.08%	98.20%
All (standard)	95.72%	88.11%	85.08%	74.04%	82.26%	98.07%
RCAug (proposed)	96.99%	88.29%	90.06%	81.92%	91.91%	97.88%

improvement than geometric and distortion transformations. Despite visually presenting good results and contours close to the gold-standard method, it is possible to note by the red and yellow arrows depicted in Fig. 9 the presence of some false negative and positive regions.

The ORCA dataset is a challenging dataset built using varied tissue samples extracted from different OSCC tumor samples. The results achieved for the ORCA dataset were slightly inferior compared to the OCDC dataset. As an external comparative result, while Martino et al. [24] obtained 56% of IOU in their best configuration, our segmentation method was six percentage points superior, achieving 62.81% of IOU in the experimental evaluation using the same test dataset (as shown in Table 3). Martino et al. [24] evaluated different color channel compositions using the RGB and HSV color spaces with four different network architectures: SegNet [44], U-Net [33], U-Net with VGG16 [45] encoder, and U-Net with ResNet50 [46] encoder. When compared with our patch-based approach, where the original images are used without resizing, our results are even better: our segmentation method was thirteen percentage points superior, achieving 69.69% of IOU (as shown in Table 2). To the best of our knowledge, our result is the first that presents a refined segmentation for OSCC-derived TMA full resolution images.

Conclusions

Data augmentation strategies become essential to train deep learning models to overcome the overfitting problem when only a few training samples are available. This paper evaluates the influence of different data augmentation transformations on the segmentation of OSCC histological images using fully convolutional neural networks. A new data augmentation strategy (RCAug) to train FCN-based segmentation methods using different image transformations is presented and evaluated. Given the input image and their corresponding label, a pipeline with a random composition of geometric, distortion, color transfer, and generative image transformations is executed on the fly. The novelty of our proposal is a new data augmentation strategy. The

study of new strategies for augmentation is crucial for deep learning methods. Good model training requires the use of huge amounts of data. In several domains, such as telecommunications, mobility, and social networks, the amount of data that can be collected is, in many situations, larger than the processing capacity. On the other hand, biomedical image data such as histological tissue samples, MRI, and CT scans are expensive to obtain in massive amounts. In addition, building a ground truth for this type of data is a time-consuming task that requires well-trained and experienced physicians and pathologists. In this way, the proposal also contributes to the deep learning algorithms applied in segmentation.

The proposal provides better precision for the delimitation of tumor regions. Clinically relevant measures adopted (e.g., F1 or accuracy) that explore the correlation of the lesion area detected by automatic and manual segmentation showed that the new strategy brought relevant contributions, as it provides a more accurate separation of tumor regions in relation to healthy tissues.

Experimental evaluations using three different datasets of H &E-stained OSCC histological images revealed that randomly composing different augmentation transformations could improve the segmentation results achieved by FCN-based segmentation methods. Moreover, considering OSCC histological images, geometric and distortion transformations have more influence on the segmentation improvement than color transformation or GAN-based generative image inpainting transformation. The experimental results also showed that using a down-sampled version of the original image could negatively impact the segmentation performance.

Two main limitations of our experiments can be listed: (i) The OCDC dataset images were originally annotated by one specialist. He has a large experience in the diagnosis of such a tumor and other oral premalignant and malignant lesions in his routine pathological examination of specimens since 2002. However, if we have more than one pathologist the interobserver variance would be addressed; (ii) the number of patients is low and image patches from the same patient were used in the training and test sets. The 15 WSI are from 15 different patients. These images were obtained from surgical samples, which are more difficult to collect on a large

scale. Although the number of images is low, the size of a WSI image is huge, and we can find great variability in the cells for the same patient, ranging from non-tumor tissue regions to tumor regions, and other cell elements.

As future directions, it will be necessary to incorporate more data augmentation transformations to evaluate which group of augmentation transformations is able to produce better segmentation results in OSCC histological images. Further investigations if the best groups of augmentation transformations for segmentation task are also the best for classification task will also be evaluated.

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Declarations

Competing Interests The authors declare no competing interests.

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